FinRL: stock selection and index enhancement

Patrick Xiong (sx2434) - Columbia STAT5398, Fall 2026

Main result. The validation-selected QQQ hybrid returned **204.45%**, versus QQQ **191.86%** and SPY **154.91%**, net of strategy trading costs from May 1, 2020 to September 30, 2025. With 20bps costs and a one-session delay it returned **198.57%**. Every headline result below uses the same frozen WRDS/CRSP data. These are developed historical strategies; no untouched holdout or statistically significant alpha is claimed.

1. Data and stock-selection method

Original notebook: 28 total cells; 20 code cells executed without errors on prepared inputs. The redundant raw-preprocessing cell was skipped. This research variant replaces the financial-ratio signal with six prior-price features after an issuer-link audit found a future-listing-dependent missingness channel. WRDS CRSP annual CIZ2025 supplies daily total returns, permanent security identifiers, dated S&P 500 membership and terminal-event records. The sample contains **705 historical stock PERMNOs, 41 decision quarters and 20,675 stock-quarter cases**.

Preparation filter counts. Point-in-time preparation excludes 110 of 815 exported stock PERMNOs that never belong at the 41 decisions. Among 27,643 positive-close, active-status stock-quarter candidates, membership removes 6,969 non-member rows and keeps 20,674. INFO's one no-start-quote member case is retained separately, giving 20,675 total cases.

Features: past momentum over 21, 63, 126 and 252 trading days (about 1, 3, 6 and 12 months), measured as log total-return growth; 63-day annualized volatility; and 126-day drawdown. Inputs use only sessions before the decision. Training-only medians and missing flags preserve incomplete rows. Financial statements, identifiers and future listing status are excluded; return labels are training targets, never predictor features.

Learning targets: a label is the realized stock return from one quarterly rebalance close to the next. Benchmark-relative models learn log(1 + stock return) minus log(1 + SPY or QQQ return) over those same dates. The absolute-target diagnostic learns stock return alone in log form. All three targets use identical stocks, features and time splits.

Training and validation: before each forecast, learn from earlier history and compare model errors on three completed quarters within a four-quarter validation span. The latest quarter ends on the forecast date, so its not-yet-observed closing outcome is excluded. Exclude training labels that overlap validation and outcomes unavailable at the applicable cutoff. Choose RF, XGB or LightGBM using lowest past validation MSE, then predict the next holding period.

Implementation: each target has 22 forecast dates. At each date, fit 11 RF seeds plus one XGB and one LightGBM: 286 estimators total. Average the 11 RF forecasts (seeds 42 and 0-9). All models use 200 trees; RF depth 6/minimum leaf 20, XGB depth 3/lr 0.05, LightGBM 15 leaves/lr 0.05. QQQ validation winners: RF 15, XGB 6, LightGBM 1.

Portfolio: quarterly top ceil(5% of eligible members), 26 stocks, equal weighted. The main strategy holds 80% QQQ and 20% actively selected stocks; the SPY hybrid substitutes SPY and its own forecasts. Active portfolios allocate 100% to actively selected stocks. Candidates remain historical S&P 500 members, not a Nasdaq-100 roster. Hybrids hold the ETF before June 1; active portfolios hold zero-yield cash until that first forecast.

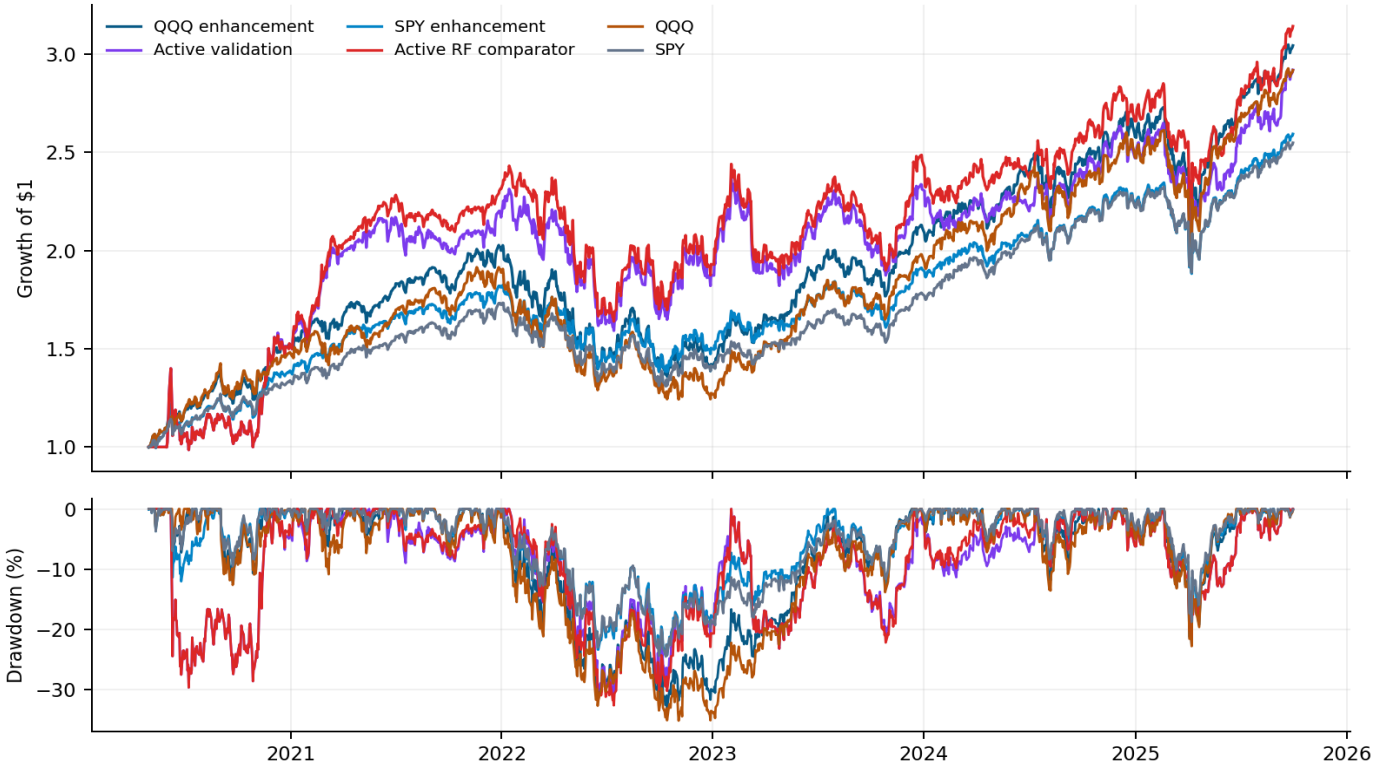
2. Performance over the same backtest period

Specification	Return	CAGR	Vol.	Sharpe*	Max DD	Turnover
QQQ enhancement	204.45%	22.82%	22.39%	1.033	-33.05%	164.09%
Active stocks, validation choice	192.04%	21.88%	29.26%	0.823	-31.20%	676.02%
SPY enhancement	159.39%	19.24%	18.85%	1.031	-24.30%	157.32%
Active stocks, RF only	214.29%	23.55%	29.38%	0.868	-32.59%	670.17%
QQQ enhancement, RF only	209.10%	23.17%	22.37%	1.046	-33.17%	163.29%
SPY	154.91%	18.86%	17.51%	1.077	-24.50%	0.00%
QQQ	191.86%	21.87%	22.99%	0.978	-35.12%	0.00%

Rows group the three headline strategies, two RF-only comparators, then ETF benchmarks; they are not sorted by return.

* Sharpe uses a zero risk-free rate and 252 sessions/year; turnover is annualized two-sided traded dollars/NAV. All strategies include 10bps per bought/sold dollar, initial funding and event exit costs. ETF benchmarks are gross buy-and-hold. Fractional shares, close fills and zero-interest cash are modeled.

3. Paths, costs and model robustness



The primary ends at $3.0445 per initial dollar versus QQQ $2.9186: **12.59 percentage points** more cumulative return, or 4.31% more terminal wealth. Its CAGR exceeds QQQ by 0.95 percentage points. The combined higher-cost/delay version still leads QQQ by **6.71 points**. This final difference does not establish persistent future alpha.

20bps + one-session delay	Return	CAGR	Max DD
QQQ enhancement	198.57%	22.38%	-33.09%
Active stocks, validation choice	166.18%	19.82%	-31.20%
Active stocks, RF only	201.03%	22.57%	-31.20%
SPY enhancement	157.21%	19.06%	-24.30%

Seed sensitivity. Keep the ensemble-derived validation family schedule fixed, replacing RF predictions by each of the 11 saved seeds only in RF-winning windows. All 11 paths exceed QQQ: mean 202.00%, range 197.47% to 209.08%, standard deviation 3.21 percentage points. No winning seed is chosen by final NAV. The primary averages forecasts, not the resulting NAVs.

Feature importance. momentum_21: 24.2%, momentum_252: 22.8%, volatility_63: 19.9%. Normalize each tree-model importance, combine each feature with its missing flag, average RF seeds, then average the validation-selected model over quarters. These are descriptive fitted-tree importances, not causal effects; all six values and window records are included.

4. Bias controls and executed evidence

Control	Evidence on the frozen source and actual execution
Historical universe	Official effective membership spells define each decision universe, independently of quote presence. All 20,675 cases and all 11,083 forecasts remain, including future exits and missing-feature cases.
End-period survivor counterexample	A September 30, 2025-only roster would delete 205 historical IDs and 3,351 rows. We retain them, including 119 securities with terminal events and 1,650 historical case rows.
Permanent identity	PERMNO identifies the security across ticker changes. Six same-ticker share-class case rows remain distinct. Ticker is only a display/tie-breaking field, never an estimator input.
Future-data counterfactual	Erase future membership records: no change over 41 decisions. Delete future raw rows before feature construction: no change on all 20,675 feature vectors.
Executed learning clocks	For each target, 858 actual fit/score/predict role records match independent row-key hashes; 286 saved estimators reproduce training medians and past-validation family choice. No recorded temporal breaches.
Adversarial tests	Poison future covariates/unavailable labels: admitted training values unchanged. Poison inference outcomes/future-status fields: input matrices identical, predictions within 1e-12. A deliberately leaky global-median negative control is detected.
Missing eligible failure label	Supervised eligibility is the start quote plus dated public-rights facts, not eventual survival or target completeness. Deliberately remove an eligible SIVB failure label: the trainer raises an error rather than silently dropping it.
Independent numerical reconstruction	Raw returns and 20,169 stock labels reconstruct with zero difference. QQQ-relative targets differ by at most 1.78e-15; a separate dollar ledger verifies 21 portfolios over 1,361 dates (max NAV difference 3.24e-14).

Failed and exited firms remain. SIVB (20 historical cases), SBNY (5) and FRC (17) retain their failure-quarter outcomes. Cash mergers and terminal returns are recognized at their source-observation date; CIZ terminal returns are used once. Verified missing-quote days receive non-executable stale portfolio marks; missing predictor quotes remain missing. Frozen held dollars cannot be sold before a tradable quote/dated settlement. Unknown held proceeds or unresolved held returns stop the replay.

Label coverage. 20,169 eligible completed-horizon labels are observed. The 506 other cases are 503 right-censored final-quarter outcomes, two unknown complete CVR outcomes (ABMD/WBA), and INFO without an executable starting quote after its February 28 terminal record. They remain forecast cases. ABMD terms are available November 4, 2022; WBA terms March 11, 2025. Their dated public-rights rule excludes new target positions before the outcome, and does not substitute invented CVR values.

Claim boundary. Conditional on historical CRSP semantics and public-term dates, the tested pipeline avoids current-survivor sample filtering and future-observation/label leakage. Original CRSP release vintages and individual membership-announcement clocks are not reconstructed. The rights rule is developed and nonexhaustive; unrestricted all-stock representativeness is not established. A read-only Claude Opus 5.5 audit (October 12, 2026) rebuilt the universe, labels, features, fit partitions and NAVs from the frozen inputs and re-ran the packaged non-training scripts. It did not refit models or certify CRSP release vintages.

5. Strategy ranking and what did not work

Target and model rule	Active stocks	80/20 hybrid	Interpretation
SPY-relative, validation choice	144.76%	159.39%	Active sleeve below SPY; hybrid slightly exceeds SPY.
QQQ-relative, validation choice	192.04%	204.45%	Three headline strategies rank QQQ hybrid > active > SPY hybrid.
QQQ-relative, RF only	214.29%	209.10%	Highest cumulative return is active RF; it is a separately declared comparator.
Absolute target, validation choice, top10%	141.82%	Not tested	Retargeting to absolute stock returns did not improve the result.

The **validation-selected** headline ordering is QQQ enhancement (204.45%) > QQQ-target active stocks (192.04%) > SPY enhancement (159.39%). The RF-only active portfolio is the highest-return displayed experiment (214.29%), but its volatility is 29.38%, Sharpe 0.868 and turnover 670.17%, versus the QQQ primary's 22.39%, 1.033 and 164.09%. Its stress return is 201.03%; the validation-selected active stress result is only 166.18%. Higher realized return does not make every active-stock rule superior.

The old free-data QQQ result (204.27%) is superseded by the WRDS replay (204.45%). All current headline results use the same sample, features, partitions, total-return conventions and costs. Source changes also replace some successor-share carry positions with event-to-cash accounting; the difference cannot be attributed solely to data quality. Benchmark/target choice and top-k research used this already-examined history. Within each run, family choice is past-only, but the overall design is a development decision.

Original workload comparison. The handout reports a 31.00% reference baseline. Reconstructing this project's own archived notebook weights with its original open-to-close proxy yields 27.16% (CAGR 4.54%, Sharpe 0.371, max DD -19.56%, volatility 15.02%, turnover 816.33%). It omits overnight returns and uses its old missing-return convention, so it is documented separately rather than inserted into the comparable CRSP close-to-close ranking.

Next evaluation. Freeze the target/model/weighting specification before a new period; test tradable fills and alternative cost assumptions. A Nasdaq-100 constituent enhancement needs a historical Nasdaq-100 roster with exits; this experiment deliberately uses historical S&P 500 candidates. Neither the old result nor this replay proves significant alpha or an untouched holdout.

6. Reproduction and submission

The folder includes source preparation, all three training targets, backtests, executed review notebook, selected stocks, compact NAV/holding files, feature importance, bias witnesses, this report and the published Medium article source text. No licensed raw data, prepared panels or fitted model weights are included. Four frozen WRDS inputs and their query/field/hash receipts reproduce the analysis. Exact commands are in README; fresh results are written to a new output directory.

```
python code/reproduce_all.py --input-dir /private/wrds/inputs \
--output-dir /private/replay
# Add --skip-absolute to replay only the two benchmark-relative targets.
```

The course destination is **submissions/Assignment1_PatrickXiong_sx2434/** in AI4Finance-Foundation/STAT-GR5398-Columbia-University, Fall-2026/Assignment1/FinRL-Trading. The author approved public course submission on October 12, 2026. The Medium article is published: FinRL stock selection: a WRDS replay of index enhancement and active portfolios. The README contains its live URL and exact reproduction commands. The accompanying course pull request proposes this folder; acceptance/merge is controlled by course maintainers.

Sources

1. Fall-2026 course requirements
2. FinRL-Trading original workload and default data
3. WRDS CRSP annual CIZ2025 daily stocks, names, delisting and S&P 500 membership exports; exact query IDs and fields in data_access.
4. CRSP daily delisting convention; ABMD/WBA dated public-term facts in data_access/dated_rights_facts.json.